Solución = soluto + solvente.

⇓

CONCENTRACIÓN

STO | SN → %m|m = g sto / 100 g SN

%m|V = g sto / 100 ml SN

%v|v = ml sto / 100 ml SN.

M = moles sto / 1L SN.

ppm = 1 parte sto / 10^6 partes SN.

STO | SVE → m = moles sto / 1 kg SVE.

Electrolitos $C^+ + A^- \xrightarrow{H_2O} C^+(aq) + A^-(aq)$

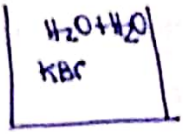
- Fuertes = disociación completa → HCl, HBr, HNO_3, $HClO_4$, Na(OH), NaCl
- Débiles = disociación parcial → HF

covalentes, hidróxidos y sales.

Fuerte: $HCl \xrightarrow{H_2O} H^+(aq) + Cl^-(aq)$ || $NaOH \xrightarrow{H_2O} Na^+ + OH^-$

Débil: $HF \xrightarrow{H_2O} H^+(aq) + F^-(aq) + HF$

7.1 KBr = soluto, 20 g. solvente = Agua 500,0 g



SN = sto + sve.

masa SN = masa sto + masa sve.

masa SN = 20 g + 500 g

masa SN = 520 g

6% m/m = g sto / 100 g SN

? x g sto → 100 g SN.

20 g sto → 520 g SN.

$x = \frac{100\,g\,SN \cdot 20\,g\,sto}{520\,g\,SN} = 3{,}85\,g\,sto.$

3,85 %m/m.

7.3 KCl 17,0 %m/m masa sto? masa sve? SN = 200 g.

17,0 % m/m = 17 g sto → 100 g SN.

34 g sto = x ← 200 g SN

masa SN = masa sto + masa sve.

masa SN − masa sto = masa sve

200 g − 34 g = masa sve

166 g = masa sve

7.5 KI = 5,00 g SN = 250 mL %m/v.

%m/v = x g sto → 100 mL SN.

5,00 g sto → 250 mL SN.

$x = \frac{5 \cdot 100\,mL}{250\,mL} = 2\,g\,sto.$

2 %m/v.

Escaneado con CamScanner

7.7) masa NaCl (sto) SN = 250 mL 2,0 M

2 M = x moles sto → 1 L = 1000 mL SN.

$x = \frac{250 \cdot 2}{1000}$ ← 250 mL SN

x = 0,5 moles. PM = PM Na + PM Cl = 23 g/mol + 35,5 g/mol = 58,5 g/mol

1 mol → 58,5 g

0,5 mol → 29,25 g = x

7.10) m ~~58,5g~~ 5,85 g NaCl → 250 g agua

moles sto → 1 kg sve. = 1000 g sve

PM NaCl = 58,5 g/mol

58,5 g → 1 mol

5,85 g → 0,1 mol

0,1 mol sto → 250 g sve

$x = \frac{1000 \cdot 0,1}{250}$ ← 1000 g sve

x = 0,4 m

7.12) 30,0 g NH_3 (sto) + 70 g H_2O (sve) ρ_{SN} = 0,982 g/mL

a) m = mol sto / 1 kg sve

b) M. = mol sto / 1 L SN.

sto NH_3 = 17 g/mol

17 g → 1 mol

30 g → x = 1,76 mol sto.

1,76 mol sto → 70 g sve

$m = x = \frac{1000 \cdot 1,76}{70}$ ← 1000 g sve

m = 25,14 molal

masa SN = masa sto + masa sve

masa SN = 30 g + 70 g

masa SN = 100 g.

$\rho_{SN} = \frac{\text{masa SN}}{\text{VL SN.}} \Rightarrow \text{VL SN} = \frac{\text{masa SN}}{\rho_{SN}}$

1,76 mol sto → 101,8 ml SN

x = 17,3 ← 1000 ml SN

$\text{VL SN} = \frac{100 \text{ g}}{0,982 \text{ g}} \text{ ml} = 101,8 \text{ ml SN}$

M = 17,3.

7.16) NaCl (aq) = SN 3,87 % m/m

SN: 75 mL → 76,9 g

V (L) SN.? 725 kg NaCl (s)

% m/m: 3,87 g sto → 100 g SN

$\rho_{SN} = \frac{76,9 \text{ g}}{75 \text{ mL}} = 1,03 \frac{\text{g}}{\text{mL}}$

725 000 g sto → x = ~~18733,85 g SN~~ $1,87 \times 10^7$ g SN.

$\text{VL SN} = \frac{\text{masa SN}}{\rho \text{ SN}} \Rightarrow \text{VL SN} = \frac{1,87 \times 10^7 \text{ g}}{1,03 \text{ g/mL}} = 1,82 \times 10^7 \text{ mL} \cdot \frac{1 \text{ L}}{1000 \text{ mL}}$

VL SN = $1,82 \times 10^4$ L.

Escaneado con CamScanner

7.23

100 mL	250 mL
40% m/m	25% m/v
ρ = 1,25 g/mL. SN	
A	B

SL > [ ] sto >

% m/m → % m/v = x g sto → 100 mL SN

40% m/m = 40 g sto → 100 g SN ⇒ 40 g sto → 80 mL SN; x 50 g sto ← 100 mL SN

mas concentrada

50% m/v = A
25% m/v = B.

$$vl\ SN = \frac{masa\ SN}{\rho SN} = \frac{100\ g\ SN}{1{,}25\ g\ SN}\ ml = 80\ mL$$

A = 50% m/v ⇒ 50 g sto → 100 mL SN → sto A = 50 g.

B = 25% m/v ⇒ 25 g sto → 100 ml SN; 62,5 g sto = x ← 250 ml SN → sto B = 62,5 g.

7.26

$KCl + H_2O \longrightarrow K^+ + Cl^-$

$CaCl_2 + H_2O \longrightarrow Ca^{2+} + 2Cl^-$

$\overset{+1}{K_3}\overset{+5}{P}\overset{-2}{O_4} + H_2O \longrightarrow 3K^+ + PO_4^{-3}$

$Fe(\overset{5}{N}\overset{-2}{O_3})_3 + H_2O \longrightarrow Fe^{3+} + 3NO_3^-$

-1

-3

7.29 $CaBr_2$ 2% m/v. → % m/v Br^- Ca^{2+}.

$1.CaBr_2 \longrightarrow 1.Ca^{2+} + 2Br^-$

2% m/v

% m/v = g sto → 100 mL SN.
2 g sto → 100 mL SN.

PM	200 g/mol	40 g/ml	+ 80 g/mol
esta	1 mol	1 mol	2 mol
masa	200 g	40 g	160 g
tengo.	2 g	0,4	1,6
	↓	↓	↓
	100 ml	100 ml	100 ml
		0,4 % m/v	1,6 % m/v.

Escaneado con CamScanner

7.33 Na_2SO_4 VSN = 1,25 dm^3 $[Na^+] = 0{,}0200\,M \rightarrow$ mol sto / 1L SL

masa?

$\Downarrow$

0,02 mol $Na^+ \rightarrow$ 1L SL

0,025 mol $Na^+ \leftarrow$ 1,25 L SL

	Na_2SO_4	$\rightarrow$	$2Na^+$	+	SO_4^{-2}
PM:	142 g/mol		23 g/mol		96 g/mol
n est:	1 mol		2 mol		1 mol
m est:	142 g		46 g		96 g
tengo:	0,0125 mol		0,025 mol		
	$\downarrow$				
	1,775 g				

Dilución $Mc \cdot Vc = Md \cdot Vd$.

$\downarrow$

%.m/v.

7.34 SL 0,750 M KOH $\longrightarrow$ 0,150 M KOH

V? 1000 cm^3

$Mc \cdot Vc = Md \cdot Vd \Rightarrow Vc = \dfrac{\overset{0,75}{Md} \cdot \overset{1000\,cm^3}{Vd}}{\underset{0,15}{Mc}}$

7.39 0,5 L Vd $\longrightarrow$ 5 cm^3 (ml) = Vc Cc $\rho = 1{,}84\,\frac{g}{cm^3}$ 98 %.m/m.

Cd [%.m/v]

$Vd\,Md = Mc\,Vc$.

Vd: L; Md: %.m/v; Mc: %.m/m; Vc: L

%.m/m $\xrightarrow{\rho}$ %.m/v.

Escaneado con CamScanner